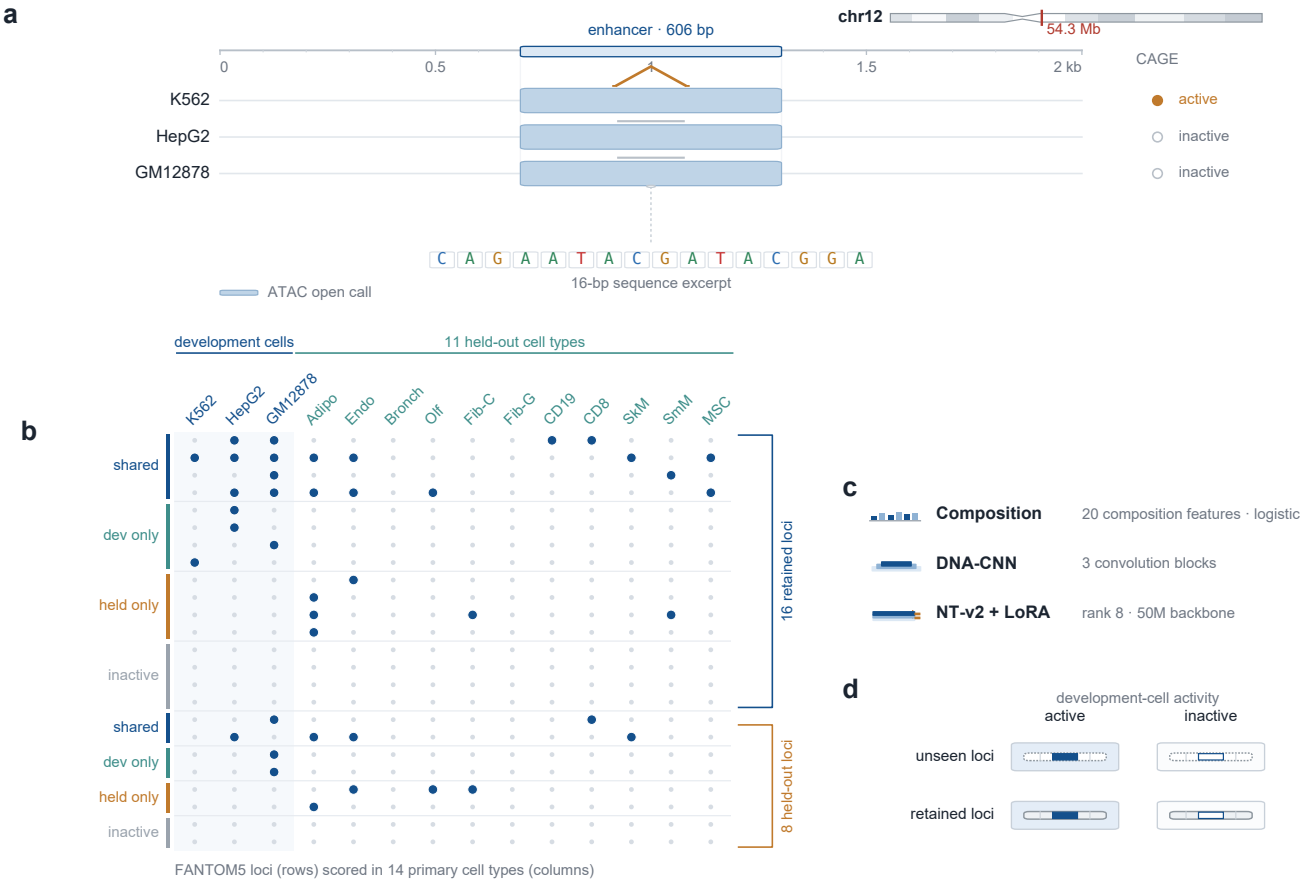

a
chr12
54.3 Mb
enhancer · 606 bp
0
0.5
1.5
2 kb
CAGE
K562
HepG2
GM12878
active
inactive
inactive
C A G A A T A C G A T A C G G A
16-bp sequence excerpt
ATAC open call
b
development cells
11 held-out cell types
K562
HepG2
GM12878
Adipo
Endo
Bronch
Olf
Fib-C
Fib-G
CD19
CD8
SkM
SmM
MSC
shared
dev only
held only
inactive
shared
dev only
held only
inactive
16 retained loci
8 held-out loci
FANTOM5 loci (rows) scored in 14 primary cell types (columns)
c
Composition
20 composition features · logistic
DNA-CNN
3 convolution blocks
NT-v2 + LoRA
rank 8 · 50M backbone
d
development-cell activity
active
inactive
unseen loci
retained loci